Network Layer: Data Plane Formula Sheet (Ch. 4)

1. Router Buffer Sizing (Rules of Thumb)

These are not strict laws but are important guidelines for engineering router buffer sizes.

Original Rule of Thumb (for a single flow)

This guideline suggests that to achieve full link utilization, the amount of buffering in a router should be equal to the link's bandwidth-delay product.

$$\text{Buffering} = RTT \cdot C$$

RTT: The average Round-Trip Time of flows passing through the link (e.g., 250 ms).
C: The capacity (bandwidth) of the link (e.g., 10 Gbps).
Example: For a 10 Gbps link and a 250 ms RTT, the required buffer size is: 0.25 sec $\times$ 10 Gbps = 2.5 Gbits.

More Recent Recommendation (for N flows)

For a link shared by a large number of TCP flows, the required amount of buffering is significantly less than the original rule of thumb suggests.

$$\text{Buffering} = \frac{RTT \cdot C}{\sqrt{N}}$$

N: The number of independent TCP flows sharing the link.
How to Use: This formula shows that as the number of flows increases, the statistical multiplexing effect reduces the amount of per-flow buffering needed to absorb bursts and maintain high link utilization.

2. Weighted Fair Queuing (WFQ)

WFQ is a scheduling discipline that provides a minimum bandwidth guarantee to different classes of traffic.

Bandwidth Allocation for a Class

Calculates the fraction of the link's total bandwidth that is guaranteed to a specific traffic class.

$$\text{Bandwidth for class } i = \left(\frac{w_i}{\sum_j w_j} \right) \cdot R$$

w_i**:** The integer "weight" assigned to traffic class i.
$\sum_j w_j$**:** The sum of the weights of *all* traffic classes.
R: The total capacity (rate) of the output link.
Example: A link has three classes with weights $w_1 = 50$, $w_2 = 30$, and $w_3 = 20$. The total of the weights is 100. Class 1 is guaranteed at least 50% of the link's bandwidth, Class 2 is guaranteed 30%, and Class 3 is guaranteed 20%.

3. IP Addressing and Subnetting

CIDR Notation

Classless Inter-Domain Routing is the standard way to represent IP addresses and their associated network prefix.

a.b.c.d /x

a.b.c.d: The IP address in dotted-decimal notation.
/x: The prefix length, indicating that the first **x** bits of the address are the **subnet part** (network address).
Host Part: The remaining (32 - x) bits of the address identify the specific host on that subnet.

Calculating the Number of Addresses in a Subnet

$$\text{Number of Addresses} = 2^{(32-x)}$$

x: The prefix length.
(32 - x): The number of bits in the host part of the address.
Example: For a subnet with prefix '/24', there are (32 - 24) = 8 host bits. The total number of available addresses in this block is $2^8 = 256$. (Note: In practice, the all-0s host address is the network address and the all-1s host address is the broadcast address, but the total size of the block is $2^{(32-x)}$).

IP Header Checksum

The checksum for the IPv4 header is calculated using the same 1's complement arithmetic as the UDP checksum, but it is computed **only over the IP header fields**. Since the TTL field changes at every router, the header checksum **must be recomputed at every router**.